U.G.C. NET Exam
COMPUTE
(Solved with Expla

Note: This paper contains seventy five (75) objective type questions of two (2) marks each. The candidates are required to select the most appropriate answer for each question. All questions are compulsory.

1. Which of the following is a correct predicate logic statement for "Every Natural number has one successor"?

(a) $\forall x \exists y(\text{succ}(x, y) \wedge (\exists z\, \text{succ}(x, z) \Rightarrow \text{equal}\ (y, z)))$

(b) $\forall x \exists y(\text{succ}(x, y) \vee (\exists z\, \text{succ}(x, z) \Rightarrow \text{equal}\ (y, z)))$

(c) $\exists x \forall y(\text{succ}(x, y) \wedge (\exists z\, \text{succ}(x, z) \Rightarrow \text{equal}\ (y, z)))$

(d) $\exists x \forall y(\text{succ}(x, y)$

Ans : (a) Statement says: For all element x their is element y which is successor of x, and if another element z is successor of x then y and z are equal. Every natural number has one successor. $\forall x \exists y(\text{succ}(x, y) \wedge (\exists z\, \text{succ}(x, z) \Rightarrow \text{equal (y,z)}))$

2. α-β cutoff are applied to _____.

(a) Depth first search
(b) Best first search
(c) Minimax search
(d) Breadth first search

Ans : (c) Alpha-beta pruning is a search algorithm that seeks to decrease the number of nodes that are evaluated by the min-max algorithm in its search tree.

3. Assume that each alphabet can have a value between 0 to 9 in a cryptoarithmetic problem

CROSS
+ROADS
DANGER

Which of the following statement is true?
(i) No two alphabets can have the same numeric value.
(ii) Any two alphabets may have the same numeric value.
(iii) $D = 0$
(iv) $D = 1$

(a) (i) and (iii)
(b) (i) and (iv)
(c) (ii) and (iii)
(d) (ii) and (iv)

Ans : (b) Given CROSS
ROADS
DANGER
whose solution is:
96233

62513
$\boxed{158746}$

There are two fairly obvious (but worth starting) rates which every alphabetic obeys:
(i) The mapping of letters to numbers is one-to-one.
(ii) the digit zero is not allowed to appear as the left most digit in any of the addends or the sum.
Note that you can find other possible solutions too.

4. Which of the following is not a part of an expert system shell?
(a) Knowledge Base (b) Inference Engine
(c) Explanation Facility (d) None of the above

Ans : (a) SHEELS - A shell is nothing but on expert system without knowledge base. A shell provides the developers with knowledge acquisition, inference engine, user interface, and explanation facility.
E.G. Java Expert system shell (JESS) that provides fully developed Java API for creating an expert system.

5. The Blocks World Problem in Artificial Intelligence is normally discussed to explain a ____.
(a) Search technique
(b) Planning System
(c) Constraint satisfaction system
(d) Knowledge base system

Ans : (b) The **blocks word** is one of the most famous planning domains in artificial intelligence. Imagine a set of wooden blocks of various shape and colors sitting on a table.
The goal is to build one or more vertical stacks of blocks.

6. Means-Ends Analysis process canters around the detection of difference between the current state and the goal state. Once such a difference is found, then to reduce the difference one applies ______.
(a) a forward search that can reduce the difference.
(b) a backward search that can reduce the difference.
(c) a bidirectional search that can reduce the difference.
(d) an operator that can reduce the difference.

Ans : (d) The **Means Ends analysis** process centers around the detection of differences between the current state and the goal state. Once such a difference is isolated, an operator that can reduce the difference must be found.

7. **Suppose a file of 10,000 characters is to be sent over a line at 2400 bps. Assume that the data is sent in frames. Each frame consists of 1000 characters and an overhead on 48 bits per frame. Using synchronous transmission, the total overhead time is _____.**

(a) 0.05 second (b) 0.1 second
(c) 0.2 second (d) 2.0 second

Ans : (c) For synchronous transmission:
= 10000 characters/1000 character per frame
= 10 frames required to transmit file
Now, overhead in bits is 10 × 48 = 480 bits.

Time overhead is = $\frac{480}{2400}$ = 0.2 second.

8. **Which of the following is the size of network bits and Host bits of Class A to IP address?**

(a) Network bits 7, Host bits 24
(b) Network bits 14, Host bits 16
(c) Network bits 15, Host bits 16
(d) Network bits 16, Host bits 16

Ans : (a) Classful IPv4 network structure:

Class	Leading bits	Size of network number bit field	Size of host bit field
A	0	7	24
B	10	14	16
C	110	21	8

For **class A network** bits and host bits are 7 bits and 24 bit respectively.

9. **Which of the following field of the TCP header tells how many bytes may be sent starting at the byte acknowledged?**

(a) TCP header length
(b) Window size
(c) Acknowledgement number
(d) urgent pointer

Ans : (b) 1. The **Window Size** field tells how many bytes will be sent starting at the byte acknowledged.
2. TCP Header field gets its name from the fact that it is also the offset from the start of the TCP segment to the actual data.

10. **Which of the following is a bit rate of an 8-PSK signal having 2500 Hz bandwidth?**

(a) 25000 bps (b) 5000 bps
(c) 7500 bps (d) 20000 bps

Ans : (c) Since $8=2^3$ bit rate is 3 times of baud rate in 8 PSK.
So, bit rate = 3 × 2500 = 7500 bps

11. **Match the following**

List- I
A. UDP

B. OSPF
C. SMTP
D. HDLC

List- I

1. Message Transfer Protocol
B. Bit-oriented Protocol
C. Interior Gateway Routing Protocol
D. Connectionless Transport Protocol

Code:

	A	B	C	D
(a)	3	4	2	1
(b)	4	3	2	1
(c)	4	3	1	2
(d)	3	4	1	2

Ans : (c) 1. UDP (User Datagram Protocol) is a connectionless transport layer protocol.
2. OSPF (Open Shortest Path First) is a routing protocol for internet protocol (IP) networks. OSPF is perhaps the most widely used interior gateway protocol (IGP) in large enterprise networks.
3. SMTP (Simple Mail Transfer Protocol) is a TCP/ IP protocol used in sending an e-mail.
4. HDLC (High Level Data Link Control) is a bit data link layer protocol.

12. Given the IP address 201.14.65 and the subnet mask 255.255.255.224. What is the subnet address?

(a) 201.14.78.32 (b) 201.14.78.64
(c) 201.14.78.65 (d) 201.14.78.224

Ans : (b) To find subnet id from subnet mask and given IP address, we do a bitwise ANDing operation. Therefore, given IP address:

$(201.14.79.65)_{10} = (11001001.00001110.01001110.01000001)_2$

Subnet mask is:

$(255.255.255.244)_{10} = (11111111.11111111.11111111.11100000)_2$

After ANDing, we get subnet id is:

$(201.14.78.64)_{10} = (11001001.00001110.01001110.01000000)_2$

13. If an artificial variable is present in the basic variable of optimal simplex table then the solution is

(a) Alternative solution
(b) Infeasible solution
(c) Unbounded solution
(d) Degenerate solution

Ans : (b) If an artificial variable is present in the "basic variable" of optimal simplex table then the solution is infeasible else if condition is false. It may be degenerate or unbounded.

14. An optimal assignment requires that the minimum number of horizontal and vertical lines that can be drawn to cover all zeros be equal to the number of

(a) rows or columns
(b) rows + columns
(c) rows + columns - 1
(d) rows + columns +1

Ans : (a) If number of lines are less than row or column then we again subtract the smallest uncovered number to obtain more number of zeroes so that number of lines covering all zeroes equal to number of rows and columns only then assignments can be made.

15. Which of the following is the minimum cost for an assignment problem given below?

		A	B	C	D
Workers	I	5	3	2	8
	II	7	9	2	6
	III	6	4	5	7
	IV	5	7	7	8

(a) 13 (b) 16
(c) 17 (d) *18*

Ans : (c) Worker (*i*) charges minimum for job C, assign him with cost 2, **worker (*ii*)** charges minimum for job C with cost 2, but C is already assigned; assign him D with cost 6.

Worker (*iii*) charges minimum for job B, assign him with cost 4. **Worker (*iv*)** is assigned with pending job A with cost 5.

Therefore, minimum cost is

$2 + 6 + 4 + 5 = 17$

16. Assume, *L* is regular language. Let statements S_1 and S_2 be defined as:

S_1: SQRT(L) = {x| for some *y* with $|y| = |x|^2$, $xy \in L$.}

S_2: LOG(L) = {x| for some *y* with $|y| = 2^{|x|}$, $xy \in L$.}

Which of the following true?

(a) S_1 is correct and S_2 is not correct.
(b) Both S_1 and S_2 are correct.
(c) Both S_1 and S_2 are not correct.
(d) S_1 is not correct ans S_2 correct.

Ans : (b) Regular languages are closed under square root, half, log operations.

17. A regular grammar for the language *L* = {*a*n *b*m|*n* even and *m* even is given by

(a) $S \to aSb \mid S_1; S_1 \to bS_1a \mid \lambda$
(b) $S \to aab \mid S_1; S_1 \to bSb \mid \lambda$
(c) $S \to aSb \mid S_1; S_1 \to S_1ab \mid \lambda$
(d) $S \to aaS \mid S_1; S_1 \to bbS_1 \mid \lambda$

Ans : (d) Given language is:

$L = \{a^n b^m \mid n \text{ is even and } m \text{ is even}\}$

Grammar for *L* is:

$S \to aaS \mid S_1 \mid \lambda$

$S_1 \to bbS_1 \mid \lambda$

Regular expression is $L = (aa)^*(bb)^*$.

18. Given the following production of a grammar.

$S \to aA \mid aBB;$

$A \to aaA \mid \lambda;$

$B \to bB \mid bbC;$

$C \to B$

Which of the following is true?

(a) The language corresponding to the given grammar is a set of even number of a's.

(b) The language corresponding to the given grammar is a set of odd number of a's.

(c) The language corresponding to the given grammar is a set of even number of a's followed by odd number to b's.

(d) The language corresponding to the given grammar is a set of odd number of a's followed by even number of b's.

Ans : (b) Given grammar is

$S \to aA \mid \boxed{aBB}$ useless production

$A \to aaA \mid \lambda$

$\boxed{B \to bB \mid bbC}$ useless production

$\boxed{C \to B}$ useless production

regular expression is

$L = a(aa)^*$

The language corresponding to the given grammar is a set of odd number of a's.

19. The language accepted by the non-deterministic pushdown automaton.

$M = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, b, z\}, \delta\, q_0, z, \{q_2\})$ with transition

$\delta\ (q_0, a, z) = \{(q_1, a)\ (q2\ \lambda)\}$;

$\delta\ (q_1, ba) = \{(q_1, b)\}$

$\delta\ (q_0, b, b) = \{(q_1\ b)\}$

$\delta\ (q_1, a, b) = \{(q_2, \lambda)\}$;

is

(a) $L(abb^*a)$

(b) $\{a\} \cup L(abb^*a)$

(c) $L(ab^*a)$

(d) $\{a\} \cup L(ab^*a)$

Ans : (d) Given states q_0, q_1, q_2 with q_0 and q_2 are starting state and final state respectively alphabet $\sum(a,b)$.

Language accepted by NPDA is

$L = \{a\} \cup \{ab^*a\}$

20. The language $L = \{a^n\, b^n\, a^m\, b^m \mid n \geq 0, n \geq 0\}$ is

(a) Context free but not linear.

(b) Context free and linear

(c) Not Context free and not linear

(d) Not Context free but linear

Ans : (a) Given language is

$L = \{a^n\, b^n\, a^m\, b^m \mid n \geq 0, m \geq 0\}$

Grammar for L is

$S \to AB$

$A \to aAb \mid \in$

$B \to aBb \mid \in$

Therefore, grammar is context-free but not linear since comparison are done here.

21. Assume statements S_1 and S_2 defined as:

S_1: L_2 - L_1 si recursive enumerable where L_1 and L_2 are recursive and recursive enumerable respectively.

S_2: The set of all Turning machines is countable.

Which of the following is true?

(a) S_1 is correct and S_2 are correct.

(b) Both S_1 and S_2 are correct.

(c) Both S_1 and S_2 are not correct.

(d) S_1 is correct and S_2 are are correct.

Ans : (b) 1. Recursive Language are closed under **complement property.** So, statement (*i*) is true.
2. The set of all turing machine is **countable** is also true.

22. Non-deterministic pushdown automaton that accepts the language generated by the grammar.
$S \rightarrow$ aSS|ab is

(a) $\delta(q_0, \lambda, z) = \{(q_1, z)\};$
$\delta(q_0, a, S) = \{(q_1, SS)\}; (q_1, B)\}$
$\delta(q_0, b, B) = \{(q_1, \lambda)\}$
$\delta(q_1, \lambda, z) = \{(q_1, \lambda)\}$

(b) $\delta(q_0, \lambda, z) = \{(q_1, Sz)\};$
$\delta(q_0, a, S) = \{(q_1, SS)\}; (q_1, B)\}$
$\delta(q_0, b, B) = \{(q_1, \lambda)\}$
$\delta(q_1, X, z) = \{(q_f, \lambda)\}$

(c) $\delta(q_0, \lambda, z) = \{(q_1, Sz)\};$
$\delta(q_0, a, S) = \{(q_1, S)\}, (q_1, B)\}$
$\delta(q_0, b, \lambda) = \{(q_1, B)\},$
$\delta(q_1, \lambda, z) = \{q_f, \lambda)\}$

(d) $\delta(q_1, X, z) = \{(q_f, z)\};$
$\delta(q_0, a, S) = \{(q_1, SS)\}, (q_1, B)\}$
$\delta(q_0, b, \lambda) = \{(q_1, B)\},$
$\delta(q_1, \lambda, z) = \{q_f, \lambda)\}$

Ans : (b) Given grammar is
$S \rightarrow aSS \mid ab$
Non-deterministic push down automata is:
$\delta(q_0, \lambda, z) = \{(q_1, Sz)\};$
$\delta(q_0, a, S) = \{(q_1, SS)\}; (q_1, B)\}$
$\delta(q_0, b, B) = \{(q_1, \lambda)\}$
$\delta(q_1, X, z) = \{(q_f, \lambda)\}$

23. Match the following

List-I
A. Dangling pointer
B. Page fault
C. List representation
D. Toss immediate

List-II
1. Buffer replacement policy
2. Variable length records
3. Object identifier

4. Pointer swizzing

Codes:

	A	**B**	**C**	**D**
(a)	3	4	2	1
(b)	4	3	2	1
(c)	4	3	1	2
(d)	3	4	1	2

Ans : (a) 1. Dangling pointers arise object destruction, when an object that has an incoming reference is deleted or deallocated, without modifying the value of the pointer, so that the pointer still points to the memory location of the deallocated memory.

2. Pointer swizzling at page fault time is a novel address translation mechanism that exploits conventional address translation hardware.

3. List representation is used to variable length records.

4. Toss immediate is a buffer replacement policy.

24. _____ Constraints ensure that a value that appears in one relation for a given set of attributes also appears for a certain set of attributes in another relation.

(a) **Logical Integrity**

(b) **Referential Integrity**

(c) **Domain Integrity**

(d) **Data Integrity**

Ans : (b) A value that appears in one relation for a given set of attributes also appears for a certain set of attributes in another relation. This is called attributes in another relation. This is called **referential integrity,** which is expressed in terms of foreign key.

1. Domain integrity specifies that all columns in a relational database must be declared upon a defined domain.

2. Data integrity is a fundamental component of information security.

25. The SQL expression

Select distinct *T*, branch_name from branch *T*, branch *S*

where *T*, assets > *S*. assets and *S*.branch_city = "Mumbai" finds the name of

(a) All branches that have greater assets than some branch located in Mumbai.

(b) All branches that have greater assets than all branches in Mumbai

(c) The branch that has greatest assets in Mumbai.

(d) Any branch that has greater assets than any branch Mumbai.

Ans : (a) Find all branches that have greater assets than some branch located in Mumbai:

Select branch_name
From branch
Where assets > Some (Select assets
                    From branch
                    Where_city = "Mumbai")

Or
Select distinct T. branch_name
From branch as T, branch as S
Where T. assets > S. assets and
S. branch_city = "Mumbai"

26. Let *A* be the set comfortable house given as

$$A=\left\{\frac{x_1}{0.8},\frac{x_2}{0.9},\frac{x_3}{0.1},\frac{x_4}{0.7},\right\}$$

and *B* be the set of affordable houses

$$\boldsymbol{B}=\left\{\frac{x_1}{0.9},\frac{x_2}{0.8},\frac{x_3}{0.9},\frac{x_4}{0.2},\right\}$$

Then the set of comfortable and affordable houses is

(a) $\left\{\frac{x_1}{0.8},\frac{x_2}{0.8},\frac{x_3}{0.1},\frac{x_4}{0.2},\right\}$

(b) $\left\{\frac{x_1}{0.9},\frac{x_2}{0.9},\frac{x_3}{0.6},\frac{x_4}{0.7},\right\}$

(c) $\left\{\frac{x_1}{0.8},\frac{x_2}{0.8},\frac{x_3}{0.6},\frac{x_4}{0.7},\right\}$

(d) $\left\{\frac{x_1}{0.7},\frac{x_2}{0.7},\frac{x_3}{0.7},\frac{x_4}{0.7},\right\}$

Ans : (a) To find the set of comfortable and affordable house, we need to find $A\cap B$.

$$A\cap B=\min[A_i,B_i]\forall i$$

So, $A\cap B$ = {min (0.8, 0.9), min (0.9, 0.8), min (0.1, 0.6), min (0.7, 0.2)}

$$=\left\{\frac{x_1}{0.8},\frac{x_2}{0.8},\frac{x_3}{0.1},\frac{x_4}{0.2},\right\}$$

27. Support of a fuzzy set

$$A=\left\{\frac{x_1}{0.2},\frac{x_2}{0.15},\frac{x_3}{0.9},\frac{x_4}{0.15},\frac{x_5}{0.15}\right\}$$

within a universal set *X* is given as

(a) $\left\{\frac{x_1}{0.15},\frac{x_2}{0.15},\frac{x_3}{0.15},\frac{x_4}{0.15},\frac{x_5}{0.15}\right\}$

(b) $\left\{\frac{x_1}{0.95},\frac{x_2}{0.95},\frac{x_3}{0.95},\frac{x_4}{0.95},\frac{x_5}{0.95}\right\}$

(c) $\{x_3,x_4\}$

(d) $\{x_1,x_2,x_3,x_4,x_5\}$

Ans : (d) Support of a **fuzzy set** *A* within a universal set *X* is

$$\{x_1,x_2,x_3,x_4,x_5\}$$

28. In a single perception, the updation rule of weight vector is given by

(a) $W(n+1) = W(n) + n[d(n) - y(n)]$
(b) $W(n+1) = W(n) - n[d(n) - y(n)]$
(c) $W(n+1) = W(n) + n[d(n) - y(n)]^{*}\times(n)$
(d) $W(n+1) = W(n) - n[d(n) - y(n)]^{*}\times(n)$

Ans : (c) The **perception learning algorithm,** which can be put into the following form:

$$w(n+1) = w(n) + \eta[d(n) - y(n)] \times x(n)$$

where, η is the step size, y is the network output and d is the desired response.

29. _______ refers to the discrepancy among a computed, Observed or measured value and the true specified or theoretically correct values.

(a) Fault (b) Failure
(c) Defect (d) Error

Ans : (d) Error is discrepancy between a computed, observed, or measured value and the true, specified or theoretically correct value or condition.

30. Which logic family dissipates the minimum power?

(a) DTL (b) TTL
(c) ECL (d) CMOS

Ans : (d) By lowering the power supply voltage CMOS chips often work a broader range of power supply voltages than other logic families. Early TTL ICs required a power supply voltage of 5 V, but CMOS could use 3 to 15 V.

31. Which of the following electronic component is not found in IC's?

(a) Diode (b) Resistor
(c) Transistor (d) Inductor

Ans : (d) 1. An **IC** is a collection of electronic component registers, transistors, capacitors etc.
2. A **transistor** is a semiconductor device just like a diode.

32. A given memory chip has 14 address pins and 8 data pins. It has the following number of locations.

(a) 2^8 (b) 2^{14}
(c) 2^6 (d) 2^{12}

Ans : (b) Memory chip has 14 address pins each pin can be 0 or 1. So, there are total 2^{14} possible addresses.

33. The instruction: MOV CL, [BX][DI] + 8 represents the _____ addressing mode.

(a) Based Relative (b) Based Indexed
(c) Indexed Relative (d) Register indexed

Ans : (b) The **based indexed addressing** mode is a combination of the based relative addressing mode and the indexed relative addressing mode. In this mode each instruction requires one base and one index register.

34. The power dissipation of a flip-flop is 3 mW. The power dissipation of a digital system with 4 flip-flops is given by

(a) 3^4 mW (b) 4^4 mW
(c) $4/3^4$ mW (d) 12 mW

Ans : (d) Given, power dissipation of one flip-flop is 3 mW. So, power dissipation of 4 flip-flops = 4 × 3 = 12 mW.

35. **An as table multivibrator using the 555 timer to generate a square wave of 5 KHz with 70% duty cycle will have**

(a) $R_a = 40.4\text{K}\Omega, R_b = 17.25K\Omega, C = 2000\text{pF}$

(b) $R_a = 17.25\text{K}\Omega, R_b = 40.4K\Omega, C = 2000\text{pF}$

(c) $R_a = 40.4\text{K}\Omega, R_b = 17.25K\Omega, C = 5000\text{pF}$

(d) $R_a = 17.25\text{K}\Omega, R_b = 40.4K\Omega, C = 5000\text{pF}$

Ans : (*) This question were marks to all. Because on solving we get

$$R_A = 1.15\,\text{Kohm}$$
$$R_B = 0.864\,\text{Kohm}$$
$$C = 0.100\,\text{uF}$$

None option matches.

36. **A binary ripple counter is required to count up to 16383. How many flip-flops are required?**

(a) 16382 (b) 8191

(c) 512 (d) 14

Ans : (d) Total number of flip-flop required to count mod n on a ripple counter is:

flip-flop = $\lceil \log_2 (n) \rceil$

Therefore,

flip-flop = $\lceil \log_2 (16383) \rceil = \lceil 13.99 \rceil$

= 14

37. **The time complexity of recurrence relation T(n) = *T*(*N/3*) + *T*(*2n/3*) + *O*(*n*) is**

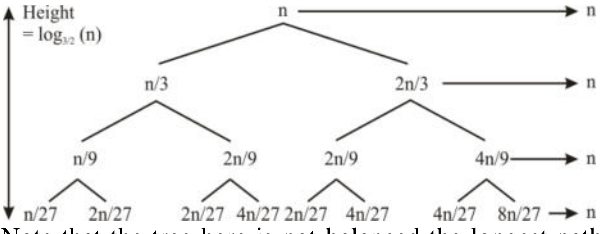
(a) $O(lg\ n)$ (b) $O(n)$

(c) $O(n\ lg\ n)$ (d) $O(n^2)$

Ans : (c) Given recurrence relation is

$$T(n) = T(n/3) + T(2n/3) + O(n)$$

Exploding out the first few levels, the recurrence is:



Note that the tree here is not balanced the longest path is the rightmost one, and its length is $\log_{3/2}(n)$.

Hence time complexity of given recurrence is $O(\log_{3/2}(n))$ or $O(n \log n)$.

38. **How many people must there be in a room before there is a 50% chance that two of them were born on the same day of the year?**

(a) At least 23 (b) At least 183

(c) At least 366 (d) At least 730

Ans : (a) 1. The **birthday problem** concerns the probability that, in a set of n randomly chosen people, some pair of them will have the same birthday.

2. By the pigeon hole principle, the probability reaches 100% when the number of people reaches 366 (since,

there are only 365 possible birthday, excluding February 29).
However, 99.9% probability is reached with just 70 people, and 50% with 23 people. These conclusions are based on the assumption that each day of the year (except February 29) is equal probable for a birthday.

$$P(A) = 1 - \frac{{}^{365}P_{23}}{(365)^{23}} = 1 - 0.4927$$

$$P(A) = 0.5072$$

39. The number of possible parenthesizations of a sequence of n matrices is

(a) $O(n)$ (b) $\Theta(n \log n)$
(c) $\Omega(n^2)$ (d) None of these

Ans : (c) Denote the number of alternative paranthesizations of a sequence of *n* matrices by p(n). Thus recurrence relation is:

$$p(n) = \begin{cases} 1 & \text{if } n = 1 \\ \sum_{k=1}^{n-1} p(k).p(n-k) & \text{if } n \geq 2 \end{cases}$$

The problem reduces to matrix multiplication problem which takes $\Omega(n^2)$.

40. The time complexity of an efficient algorithm to find the longest monotonically increasing subsequence of n numbers is

(a) $O(n)$
(b) $O(n \log n)$
(c) $O(n^2)$
(d) None of these

Ans : (b) The longest increasing subsequence problem is solvable in time O(n log n), where *n* denotes the length of the input sequence.

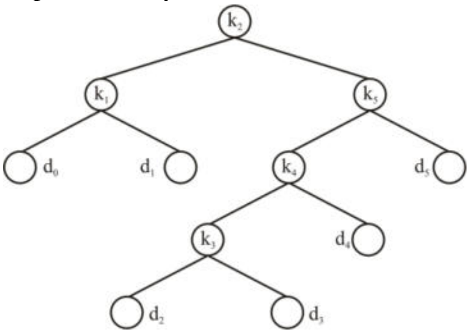
41. Given a binary search trees for a set of *n* = 5 *keys* with the following probabilities:

i	**0**	**1**	**2**	**3**	**4**	**5**
P_i	**—**	**0.15**	**0.10**	**0.10**	**0.10**	**0.20**
q_i	**0.05**	**0.10**	**0.05**	**0.05**	**0.05**	**0.10**

The expected optimal cost of the search is

(a) 2.65 (b) 2.70
(c) 2.75 (d) 2.80

Ans : (c) Optimal binary search tree is:



This tree has optional expected cost is 2.75.

42. Given 0-1 knapsack problem and fractional knapsack problem and the following statements:

S_1: 0-1 knapsack is efficiently solved using Greedy algorithm.

S_2: Fractional knapsack is efficiently solved using Dynamic programming.

Which of the following is true?

(a) S_1 is correct and S_2 is not correct.
(b) Both S_1 and S_2 are correct.
(c) Both S_1 and S_2 are not correct.
(d) S_1 is not correct and S_2 is correct.

Ans : (*) Since, greedy algorithm optimized local and dynamic algorithm optimized global.
So, greedy algorithm does not work for 0 - 1 Knapsack problems correctly 0-1 Knapsack efficiently dynamic programming but fractional Knapsack solved efficiently using greedy algorithm.

43. Equivalence class partitioning approach is used of divide the input domain into a set of equivalence classes, so that if a program works correctly for a value, then it will work correctly for all the other values in that class. This is used _______.

(a) to partition the program in the form of classes.
(b) to reduce the number of test form cases required.
(c) for designing test cases in white box testing.
(d) all of the above.

Ans : (b) Equivalence class partitioning is a software testing technique that divides the input data of a software unit into partitions of equivalent data from which test cases can be derived. An advantage of this approach is reduction in the time required for testing a software due to lesser number of test cases.

44. The failure intensity for a basic model as a function of failures experience is given as $\lambda(\mu) - \lambda_0\ [1-(\mu)/V_0)]$.

where λ_0 is the initial failure intensity at the start of the execution, μ is the average or expected number of V_0 failure at a given point in time, the quantity V_0 is the total number of failure that would occur in infinite time.

Assume that a program will experience 100 failures in infinite time, the initial failure intensity was 10 failures/CPU hr. Then the decrement of failures intensity per failure will be

(a) 10 per CPU hr.
(b) 0.1 per CPU hr.
(c) -0.1 per CPU hr.
(d) 90 per CPU hr.

Ans : (c) The decrement of failure intensity per failure is:

$$\Rightarrow \frac{d\lambda}{d\mu} = -\left(\frac{\lambda_0}{Y_0}\right) = -\frac{10}{100}$$

$\Rightarrow -0.1$ per CPU hr.

45. Improving processing efficiency or performance or restructuring of software to improve changeability is known as

(a) Corrective maintenance
(b) Perfective maintenance
(c) Adaptive maintenance
(d) Code maintenance

Ans : (b) 1. Perfective maintenance modification of a software product after delivery to improve performance or maintainability.

2. Corrective maintenance reactive modification of a software product performed after delivery to correct discovered problems.

3. Adaptive maintenance modification of a software product performed after delivery to keep a software product usable in a changed or changing environment.

46. In_____, modules A and B make use of a common data type, but perhaps perform different operations on it.

(a) Data coupling (b) Stamp coupling
(c) Control coupling (d) Contents coupling

Ans : (b) 1. Two modules are **stamp coupled** if they communicate via a passed data structure that contains more information than necessary for them to perform their functions.

2. Two modules are **data coupled** if they communicate by passing parameters.

3. Two modules are **control coupled** if they communicate using at least one "control flag".

47. Consider the following schemas:

Branch_Schema= (branch_name, assets, city)
Customer_Schema=(customer_name, street, city)
Deposit_Schema = (branch_name, account _number, customer_name, balance)
Borrow_Schema = (branch_name, loan_number, customer_name, amout)
Which of the following tuple relational calculus finds all customers who have loan amount more than Rs. 12,000?

(a) $\{t(\text{customer\_name}) | t \in \text{borrow} \wedge \ t[\text{amount} > 12000\}$
(b) $\{t | t(\text{customer\_name}) | t \in \text{borrow} \wedge \ t[\text{amount} > 12000\}$
(c) $\{t | \text{s} \ \ \in \text{borrow} \wedge (t(\text{cumtomer\_name} = \text{s} (\text{customer\_name})) \ \text{s}[\text{amount}] > 12000\}$
(d) $\{t | \text{s} \ \ \in \text{borrow} \wedge (t(\text{cumtomer\_name} \wedge \text{s} (\text{customer\_name})) \ \text{s}[\text{amount}] > 12000\}$

Ans : (c) Query: Find all customers who have loan amount more than Rs. 12000.

TRC:

$\{t | s \in \text{borrow} \wedge (t(\text{customer\_name} = s(\text{customer\_name})) \, s[\text{amount}] > 12000\}$.

48. **Match the following:**

List-I	**List-II**
A. Create	**1. The E-R Model**
B. Select	**2. Relationship Model**
C. Rectangle	**3. DDL**
D. Record	**4. DML**

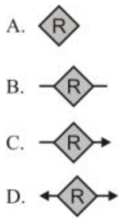
Code:

	A	**B**	**C**	**D**
(a)	3	4	1	2
(b)	4	3	2	1
(c)	4	3	1	2
(d)	3	4	2	1

Ans : (a) 1. Create used in DLL (data definition language).
2. Select used in DML (data manipulation language).
3. Rectangle used in ER diagram as relationship.
4. Record or information used in relationship model.

49. **Match the following:**

List-I



List-II

1. One to one relationship
2. Relationship
3. Many to many relationship
4. Many to one relationship

Codes:

	A	**B**	**C**	**D**
(a)	3	4	2	1
(b)	4	3	2	1
(c)	2	3	4	1
(d)	3	4	1	2

Ans : (c)

ER-diagram	**Description**
	Relationship
	Many-to-many relationship
	Many-to-one relationship
	One-to-one relationship

50. **Sixty (60) reusable components were available for an application. If only 70% of these components can be used, rest 30% would have to be developed from scratch. If average component is 100 LOC and cost of each LOC is Rs. 14, what will be the risk exposure if risk probability is 80%?**

(a) Rs. 25,200 (b) Rs. 20,160
(c) Rs. 25,160 (d) Rs. 20,400

Ans : (b) 60 reusable software components were planed. If only 70 percent can be used then 30% i.e., 18 components would have to be developed from scratch. Since the average component is 100 LOC and local data indicate that the software engineering cost for each LOC is Rs. 14 the overall cost (impact) to develop the components would be $18 \times 100 \times 14$ = Rs. 25,200
So, Risk exposure = $0.8 \times 25,200$ = Rs. 20,160.

51. Consider the following two function declarations:

(*i*) int *f*() (*ii*) int(**f*)()

Which of the following is true?

(a) Both are identical
(b) The first is a correct declaration and the second is wrong.
(c) Both are different ways of declaring pointer to a function.
(d) The first declaration is a function returning a pointer to an integer and the second is a pointer to function returning integer.

Ans : (d) int *f( ) is a function returning a pointer to an integer.
int (*f) ( ) is a pointer to function returning integer.

52. Assume that we have constructor function for both Base and Derived classes. Now consider the declaration:

main ( )
Base *P = new Derived;
In what sequence, the constructor will be executed:

(a) Derived class constructor is followed by Base class constructor.
(b) Base class constructor.
(c) Base class constructor is never called.
(d) Derived class constructor is never called.

Ans : (b) Whenever a derived class object is created, the **base class constructor** gets called first and then the **derived class constructor.**

53. What is the output of the following program?

```
#include <stdio.h>
main ( )
{
  int a, b = 0;
  static int c[10] = 2, 3, 4, 5, 6, 7, 8, 9, 0]}
  for (a = 0; a < 10; + + a)
  if ((C[a]% 2==0) b+ = c [a];
  printf("d", b);
}
```

(a) 15 (b) 25
(c) 45 (d) 20

Ans : (d) The print *f* statement will print 'b' which is sum of those values from array *C* which get divided by 2, that is 2 + 4 + 6 + 8 = 20.

54. A Program contains the following declaration and initial assignments

int i = 8, j = 5;

double x = 0.005, y = -0.01;

char c = 'c' d = 'd';.

Determine the value of the following expressions which involve the use of library functions:

abs *(i - 2*j)*; log (exp(x)); toupper (d)

(a) 2; 0.005; D (b) 1; 0.005; D

(c) 2; 0.005; E (d) 1; 0.005; e

Ans : (a) abs(i -2j) returns the absolute value = 2 log (exp(x)) = In (e^x) = 0.004.?
toupper 'D' letter to be converted to upper case.

55. MPEG involves both spatial compression and temporal compression. The spatial compression is similar to JPEG and temporal compression removes _______ frames.

(a) Temporal (b) Voice

(c) Spatial (d) Redundant

Ans : (d) MPEG (Motion Pictures Experts Group) compression is a method of compressing video. MPEG involves both **spatial compression** and **temporal compression.**
The spatial compression is similar to JPEG and temporal compression removes redundant frames.

56. If the data unit is 111111 and the divisor in 1010 in CRC method, what is the dividend at the transmission before division?

(a) 1111110000 (b) 1111111010

(c) 111111000 (d) 111111

Ans : (c) In **CRC** generating method, if divisor is n bit long then we add (n - 1) number of 0's to LSB in the data unit before division.
So, here divisor is 1010 hence 3 0's are added to LSB of data unit before division. That is 111111000.

57. If user A wants to send an encrypted message to user B. The plain text of A is encrypted with the _______ .

(a) Public Key of user A

(b) Public Key of user B

(c) Private Key of user A

(d) Private Key of user B

Ans : (b) 1. Two parties (*A* and *B*) might use **public key encryption** as follows: First *A* generate a public/private key pair. If *A* wants to send *B* an encrypted message, *A* ask *B* for *B*'s public key. *B* sends *A* the public key over an insecure network, and then *A* uses this key to encrypt a message.
2. *A* sends the encrypted message to *B*, and then *B* decrypts it by using *B*'s private key. If *A* received *B*'s key over an insecure channel, such as public network, *A* must verify with *B* that *A* has a correct copy of B's public key.

58. A _______ can forward or block packets based on the information in the network layer and transport layer header.

(a) Proxy firewall
(b) Firewall
(c) Packer fitter firewall
(d) Message digest firewall

Ans : (c) 1. A **firewall** can be used as a packet filter. It can forward or block packets based on the information in the network layer and transport layer headers: source and destination IP addresses, source and destination port addresses, and type of protocol (TCP or UDP).
2. A **packet filter firewall** is a router that uses a filtering table to decide which packets must be discarded (not forwarded).

59. Which of the following graphics devices are known as active graphics devices?
(i) Alphanumeric devices
(ii) Thumb wheels
(iii) Digitizers
(iv) Joy stick

(a) (i) and (ii)
(b) (iii) and (iv)
(c) (i), (ii) and (iii)
(d) (i), (ii), (iii) and (iv)

Ans : (d) 1. Active graphics devices has dynamic nature control, 2-way communications, interaction with high bandwidth user.
2. These are modern applications with 2-D, 3-D transformations.
3. All given devices are active graphics devices.

60. A diametric projection is said to be trimetric projection when
(i) two of the three foreshortening factors are equal and third is arbitrary.
(ii) all of the three foreshortening factors are equal.
(iii) all of the three foreshortening factors are arbitrary.
Which of the above is true?

(a) (i) and (ii) (b) (ii) and (iii)
(c) (i) only (d) (iii) only

Ans : (c) A **diametric projection** is a special case of trimetric projection where two of the three foreshortening factors are equal and third arbitrary. Therefore only statement (*i*) is true.

61. Which of the following is / are fundamental method(s) of antialiasing?
(i) Increase of sample rate.
(ii) Treating a pixel as a finite area rather than as a point.
(iii) Decrease of sample rate.

(a) (i) and (ii) (b) (ii) and (iii)
(c) (i) only (d) (ii) only

Ans : (a) 1. Antialiasing is the smoothing of the image or sound roughness caused by aliasing.

2. Antialiasing is the application of techniques that reduce or eliminate aliasing. Increasing the sample rate and treating a pixel as a finite area rather than as a point are fundamental methods of antialiasing.

62. The two color systems - the HSV and HLS are

(a) Hue, Saturation, Value and Hue, Lightness, Saturation.

(b) High, Standard, Value and High, Lightness, Saturation.

(c) Highly, Standard, Value and Highly, Lightened, Saturation.

(d) Highly, Standard, Value and Hue, Lightness, Saturation.

Ans : (a) HSL stands for **hue, saturation** and **lightness** (or luminosity) and is also often called HLS. HSV Stands for Hue, saturation and value also called HSB (B = brightness).

63. The parametric representation of the line segment between the position vectors $P_1(2,3)$ and $P_2(5,4)$ is given as

(a) $x(t)=2+7t, y(t)=0\le t\le\infty 3+7t$

(b) $x(t)=2+10t, y(t)=0\le t\le 1\ 3+12t$

(c) $x(t)=2+3t, 0\le t\le 1\ y(t)=3+t$

(d) $t(x,y)=14t\ 0\le t\le 1$

Ans : (c) Given two points (x_1, y_1) and (x_2, y_2) the point (x, y) is on the line determined by (x_1, x_2) and (x_2, y_2) if and only if there is a real number t such that

$$x=(1-t)x_1+t_1x_2$$

$$y=(1-t)y_1+ty_2$$

Here, $(x_1, y_1) = (2, 3)$ and $(x_2, y_2) = (5, 4)$.

Therefore,

$$x=(1-t)2+t\cdot 5=3t+2$$

$$y=(1-t)3+t\cdot 4=t+3$$

64. Consider the following transformation matrix for rotation (clockwise):

$$[T]=\begin{bmatrix} \text{COS}\theta & \text{Sin}\,\theta & 0 & 0 \\ -\text{Sin}\,\theta & \text{COS}\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

This matrix rotates an object by an angle 9 about:

(a) X-axis (b) Y-axis

(c) Z-axis (d) All of these

Ans : (c) Rotations in space are more complex, because we can either rotate about the x-axis, the y-axis, or the z-axis. When rotating about the z-axis, only coordinates of x and y will change and z coordinate will be the same. In effect, it is exactly a rotation about the origin in the xy-plane.

Therefore, the rotation equation is:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \\ 1' \end{bmatrix}$$

With this set of equations, letting θ be 90 degree rotates (1, 0, 0) to (0, 1, 0) and (0, 1, 0) to (-1, 0, 0). Therefore, the x-axis rotates to the y-axis and the y-axis rotates to the negative effect of rotating about the z-axis 90 degree.

65. Consider the following sequence of operations:

(i) Pointer p_1 is set to point at a new heap-dynamic variable.

(ii) Pointer p_2 is assigned p_1's value.

(iii) The heap dynamic variable pointed to by p_1 is explicitly de-allocated, but p_2 is not changed by the operation.

This situation leads to which of the following:

(a) p_1 becomes a dangling pointer

(b) p_2 becomes a dangling pointer

(c) Both p_1 and p_2 are now dangling pointers

(d) Neither p_1 nor p_2 is now a dangling pointer

Ans : (b) A **dangling pointer** is a pointer that contains the address of a dynamically allocated memory that has been deallocated.

So, P_2 is now a dangling pointer. If the deallocation operation did not change P_1, both P_1 and P_2 would be dangling.

66. The programming languages C and C++ are not strongly typed languages because:

(a) Both C and C++ allow functions for which parameters are not type checked.

(b) Both C and C++ allow functions for which parameters are type checked.

(c) Both C and C++ allow functions for which parameters are not type checked and also the union types in these languages are not type checked.

(d) Union types in these (C and C++) languages are not type checked.

Ans : (c) 1. In contrast, a language is weakly typed if type confusion can occur. Silently (undetected), and eventually case errors that are difficult to localize.
2. Also, C and C++ are considered weakly typed since, due to typecasting, one can interpret a field of a structure that was an integer as a pointer.
3. C and C++ are not strongly typed language because both allow functions for which parameters is not type checked. The union types of these checked. The union types of these languages are not type checked.

67. The tracing model in Prolog describes program execution in terms of certain events. These events are

(a) call and exit

(b) call and fail

(c) call, exit and redo

(d) call, exit, redo and fail

Ans : (d) The **Tracing Model** describes the execution of prolog program in terms of four kinds of events that occurs. These are Call, Exit, Redo and Fail. The debugging aids tell us about when event of these four kinds occur in the execution of our programs. These events will take place for all of the various goals that prolog considers during the execution.

68. Which of the following statements is not correct with reference to distributed systems?

(a) Distributed system represents a global view of the network and considers it as a virtual uni-processor system by controlling and managing resources across the network on all the sites.

(b) Distributed system is built on bare machine, not an add-on to the existing software.

(c) In a distributed system, kernel provides smallest possible set of services on which other services are built. This kernel is called microkernel. Open serves provide other services and access to shared resources.

(d) In a distributed system, if a user wants to run the program on other nodes or share the resources on remote sites due to their beneficial aspects, user has to log on to that site.

Ans : (d) A **distributed system** is a model in which components located on networked computers communicate and coordinate their actions by passing messages. The components interact with each other in order to achieve a common goal.

69. A system contains 10 units of resource of same type. The resource requirement and current allocation of these resources for three processes P, Q and R are as follows:

	P	**Q**	**R**
Maximum requirement	**8**	**7**	**5**
Current allocation	**4**	**1**	**5**

Now, consider the following resource requests:
(i) *P* makes a request for 2 resource units.
(ii) *Q* makes request for 2 resource units.
(iii)*R* makes a request of 2 resource units.
For a safe state, which of the following options must be satisfied?

(a) Only request (i) (b) Only request (ii)
(c) Only request (iii) (d) Request (i) and (ii)

Ans : (c)

Process	**Max Need**	**Allocated**	**Available**	**Remaining Need**
P	8	4	2	4
Q	7	1		6
R	5	3		2
		8		

Available = 10 - 8 = 2
Only request of *R* (of 2 resources) must be satisfied so that it can be completed and release all 5 resources which in turn will ensure the execution of *P* then *Q* that is a safe sequence.

70. Consider the following set of processes with the length of CPU burst time in milliseconds (ms):

Process	**A**	**B**	**C**	**D**	**E**
Burst time	**6**	**1**	**2**	**1**	**5**
Priority	**3**	**1**	**3**	**4**	**2**

Assume that processes are stored in ready queue in following order:
A - B - C - D - E
Using round robin scheduling with time slice of 1 ms. the average turn around time is

(a) 8.4 ms (b) 12.4 ms
(c) 9.2 ms (d) 9.4 ms

Ans : (a) Gantt chart is

B	E	A	C	D	E	A	C	E	A	E	A	E	A	A

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Therefore, average TAT $=\dfrac{15+1+8+5+13}{5}$

$=\dfrac{42}{5}=8.4$ milliseconds

71. Consider a main memory with 3 page frames for the following page reference string:

5, 4, 3, 2, 1, 4, 3, 5, 4, 3, 4, 1, 4

Assuming that the execution of process is initiated after loading page 5 in memory, the number of page faults in FIFO and second chance replacement respectively are

(a) 8 and 9 (b) 10 and 11

(c) 7 and 9 (d) 9 and 8

Ans : (d) Using **FIFO** (First in First out) replacement algorithm (given page 5 already in memory, so, there is no page fault for 5 in first).

initial 5 4 3 2 1 4 3 5 4 3 4 1 4

5		5	5	2	2	2	3	3				3	4
		4	4	4	1	1	1	3				5	5
			3	3	3	4	4	4				1	1

Therefore, total number of page fault in FIFO algorithm is 9 and replacement is 8.

72. Which of the following shell scripts will produce the output "my first script"?

(a) for *i* in my first script {echo - *i* $*i*}

(b) for my first script; do echo - n; done

(c) for *i* in my first script; do echo - *i* $*i*; done

(d) for n in my first script; do echo - *i* $*i*; done

Ans : (c) For loop control variable *i* which will be assigned the value after it and do apply for loop to print it should be enclosed in do .... done block.

Therefore, echo - *i* display the current content of *i*.

73. The portion of Windows 2000 operating system which is not portable is

(a) processor management

(b) user interface

(c) device management

(d) virtual memory management

Ans : (d) Each processor that windows 2000 supports implements virtual memory through hardware differently; therefore, the portion of windows 2000 that directly interfaces with virtual memory hardware is not portable and must be recorded when moving to another platform to minimize problem.

74. Match the following for Windows Operating System:

List - I

A. Hardware abstraction layer

B. Kernel

C. Executive

D. Win32 subsystem

List - II

1. **Starting all processes, emulation of different operating systems, security functions, transform character based applications to graphical representation.**
2. **Export a virtual memory interface, support for symmetric multiprocessing, administration, details of mapping memory, configuring I/O buses, setting up DMA.**
3. **Thread scheduling, interrupt and exception handling, recovery after power failure.**
4. **Object manager, virtual memory manager, process manager, plug-and-play and power manager.**

Codes:

	A	**B**	**C**	**D**
(a)	1	3	2	4
(b)	4	3	2	1
(c)	2	3	4	1
(d)	3	2	1	4

Ans : (c) There, (a)-(*ii*), (b)-(*iii*), (c)-(*iv*), (d)-(*i*)

75. Which of the following statements is not true for UNIX Operating System?

(a) Major strength of UNIX Operating System is its open standards which enabled large number of organizations ranging from academicia to industries to participate in its development.

(b) UNIX kernel uses modules with well specified interfaces and provides advantages like simplified testing and maintenance of kernel. It has better extensibility as the kernel is not monolithic.

(c) UNIX is kernel based operating system with two main components viz. process management subsystem and file management subsystem.

(d) All devices are represented as files which simplify the management of I/O devices and files. The directions structure used is directed acyclic graph.

Ans : (b) Use of **Kernel modules** with well specified interfaces provides several advantages. Existence of the module interface simplifies testing and maintenance of the kernel. An individual module can be modified to provide new functionalities or enhance existing once. This feature overcomes the poor extensibility typically associated with monolithic kernels.